

Fibonacci Numbers in an Interview Question

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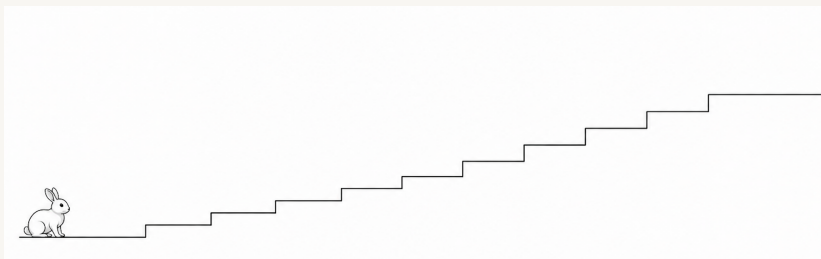
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Abstract

When people say Fibonacci numbers are everywhere, its because its true. Here I will discuss a combinatorial argument for a nice Fibonacci identity I found while doing a practice interview question.

The following is the question of interest and was taken from Xinfeng Zhou's book 'A practical guide to quantitative finance interviews'.

Hopping rabbit. A rabbit sits at the bottom of a staircase with n stairs. The rabbit can hop up only one or two stairs at a time. How many different ways are there for the rabbit to ascend to the top of the stairs?



Before we start going through a solution I encourage you to try it yourself. What will be presented is a combinatorial argument for a general solution, and then (a much easier) recursive relationship which is then solved by the Fibonacci numbers, giving a nice identity.

Combinatorial argument

We will break up the cases into n even and n odd. Let's start with n even.

Let $n = 2m$ for some integer m . Whatever way the rabbit chooses to go up the stairs, it decides on a number of hops where it jumps one stair and a number of hops where it jumps two stairs. Thinking about the possible number of each type of jump we arrive at the following table of possibilities, with each column being a possible way to get to the top.

Number of hops up one stair	$2m$	$2m - 2$	$2m - 4$	$\dots$	4	2	0
Number of hops up two stairs	0	2	4	$\dots$	$m - 4$	$m - 2$	m

For m hops up one stair and m hops up two stairs, there is only one way to achieve this.

For all other columns, the rabbit chooses $2k$ hops up one stair and $m - k$ hops up two stairs. For each column and $1 \leq k \leq m - 1$, there is exactly $\binom{m+k}{2k}$ ways of doing this. This is due to there being $2k + (m - k) = m + k$ total hops, and we have to choose $2k$ of these hops to be hops that go up one stair.¹

As a result, the rabbit has exactly

$$2 + \sum_{k=1}^{m-1} \binom{m+k}{2k}$$

ways of ascending to the top of the stairs for $n = 2m$ stairs.

Now let $n = 2m + 1$. Similarly, the rabbit must decide on a number of hops where it jumps one stair and a number of hops where it jumps two stairs. We then arrive at the following table.

Number of hops up one stair	$2m + 1$	$2m - 1$	$2m - 3$	$\dots$	5	3	1
Number of hops up two stairs	0	2	4	$\dots$	$m - 4$	$m - 2$	m

For $2m + 1$ hops up one stair, there is only one way to achieve this.

For all other columns, the rabbit chooses $2k + 1$ hops up one stair and $m - k$ hops up two stairs. For each column and $0 \leq k \leq m - 1$, there is exactly $\binom{m+k+1}{2k+1}$ ways of doing this. This is due to there being $2k + 1 + (m - k) = m + k + 1$ total hops, and we have to choose $2k + 1$ of these hops to be hops that go up one stair.²

Consequently, the rabbit has exactly

$$1 + \sum_{k=0}^{m-1} \binom{m+k+1}{2k+1}$$

ways of ascending to the top of the stairs for $n = 2m + 1$ stairs.

We can summarise these results by the following. Let $f(n)$ be the number of ways of ascending to the top of n stairs, then,

$$f(2m) = 2 + \sum_{k=1}^{m-1} \binom{m+k}{2k}$$

and

$$f(2m + 1) = 1 + \sum_{k=0}^{m-1} \binom{m+k+1}{2k+1}.$$

Recursive argument

Let $f(n)$ be the number of ways of ascending to the top of n stairs. Clearly, $f(1) = 1$ and $f(2) = 2$.

The rabbit's first hop could jump up one stair or two stairs. If the rabbit chooses to jump up one stair, then the number of ways to ascend to the top of the remaining $n - 1$ stairs is $f(n - 1)$. If instead the

¹Note you have some symmetry as you could argue that we have to choose $m - k$ of these hops to be hops that go up two stairs, giving $\binom{m+k}{m-k}$ which is in fact equal to $\binom{m+k}{m-k}$, can you explain why?

²Again, you can use symmetry for the same reason as the above footnote.

rabbit chooses to jump up two stairs, then the number of ways to ascend to the top of the remaining $n - 2$ stairs is $f(n - 2)$. Hence,

$$f(n) = f(n - 1) + f(n - 2)$$

with $f(1) = 1$ and $f(2) = 2$.

Fibonacci numbers

Now for the interesting part. The solutions to the recursive argument are exactly the Fibonacci numbers.

Recall, that the Fibonacci numbers are the set of numbers F_n with $F_1 = 1$ and $F_2 = 2$ such that³,

$$F_n = F_{n-1} + F_{n-2}$$

which is exactly our $f(n)$, and so $F_n = f(n)$. In our combinatorial argument, we found a relation for these F_n , that is for $m \geq 2$,

$$F_{2m} = 2 + \sum_{k=1}^{m-1} \binom{m+k}{2k}$$

and

$$F_{2m+1} = 1 + \sum_{k=0}^{m-1} \binom{m+k+1}{2k+1}.$$

which are quite fiddly to prove, try it yourself!

This just goes to show, Fibonacci numbers always pop up even when you least expect them.

³Some authors choose $F_1 = F_2 = 1$ for the seeds, but for our question it will look nicer with these.